

FinPredict

Test Plan, Test Cases & Test Report

Version 1.0 | April 2026

1. Test Plan

1.1 Test Objectives

- Verify all API endpoints return correct HTTP status codes and response bodies
- Verify prediction probability is always between 0.0 and 1.0
- Verify preprocessing transforms inputs correctly and consistently
- Verify invalid inputs are rejected with appropriate error codes
- Verify the frontend renders without errors and predictions display correctly
- Verify system meets the <200ms latency business requirement

1.2 Test Scope

In Scope	Out of Scope
Backend API unit tests (pytest)	Load testing / performance testing
Preprocessor unit tests	Security penetration testing
Health endpoint tests	Cross-browser compatibility testing
Input validation tests	Mobile device testing
Frontend end-to-end manual tests	Third-party service integration tests

1.3 Test Environment

Component	Version/Details
Operating System	Ubuntu 24.04 LTS
Python	3.12.3
Test Framework	pytest 7.4.3 with anyio plugin
Test Client	FastAPI TestClient (httpx-based)
Browser for Manual Tests	Google Chrome 147
MLflow Server	Running at http://localhost:5000
Backend Server	Running at http://localhost:8000

1.4 Acceptance Criteria

Criteria	Target	Status
All unit tests pass	10/10 tests passing	MET

API returns correct HTTP codes	200, 422, 500 as specified	MET
Prediction probability range	Always $0.0 \leq \text{prob} \leq 1.0$	MET
Invalid inputs rejected	422 for missing/wrong fields	MET
Inference latency	< 200ms per prediction	MET (avg < 50ms)
Frontend renders correctly	No JavaScript console errors	MET
High credit score → approval	credit_score=820 → approved=true	MET
Low credit score → rejection	credit_score=320 + high debt → approved=false	MET

2. Test Cases

2.1 Unit Test Cases — Backend API

ID	Module	Test Name	Input	Expected Output	Priority
TC01	Health	test_health_returns_200	GET /health	200, {status:healthy}	High
TC02	Health	test_ready_returns_200	GET /ready	200, {status:ready}	High
TC03	Predict	test_predict_returns_200	Valid 24-field payload	200 OK	Critical
TC04	Predict	test_predict_response_has_required_fields	Valid payload	6 fields present	Critical
TC05	Predict	test_predict_probability_between_0_and_1	Valid payload	$0.0 \leq \text{prob} \leq 1.0$	Critical
TC06	Predict	test_predict_invalid_payload_returns_422	gender='Unknown'	422 Unprocessable	High
TC07	Preprocessor	test_preprocess_returns_dataframe	LoanApplication object	pandas DataFrame	High
TC08	Preprocessor	test_preprocess_has_one_row	LoanApplication object	shape[0] == 1	Medium
TC09	Preprocessor	test_debt_to_income_ratio	debt=8000, income=85000	0.094117...	Medium
TC10	Preprocessor	test_gender_encoded_correctly	gender=Male	Gender column == 1	Medium

2.2 Manual Test Cases — Frontend

ID	Test Description	Steps	Expected Result
MT01	Homepage loads correctly	Open http://localhost:3002	Hero, features, CTA button visible
MT02	Navigate to Apply page	Click Apply in navbar	Loan form with 4 sections visible
MT03	Form accepts all input types	Fill all 24 fields	No focus loss, values retained
MT04	Approved prediction displays	Enter high income (250000), credit score 800, submit	Green Approved card with probability > 75%
MT05	Rejected prediction displays	Enter income 15000, credit score 320, debt 50000, submit	Red Rejected card with recommendation

MT06	Pipeline page loads	Click Pipeline in navbar	7 task statuses and tool links visible
MT07	User Manual page loads	Click User Manual in navbar	All 10 sections visible
MT08	Empty form submission	Click submit without filling fields	Error message or 422 error visible

2.3 Integration Test Cases

ID	Test Description	Expected Result
IT01	Frontend → Backend prediction	Form submission reaches /api/v1/predict and result renders
IT02	Backend → MLflow model loading	Backend starts and logs: Model loaded from MLflow
IT03	Backend → Prometheus metrics	GET /metrics returns prediction counters after predictions
IT04	Prometheus → Grafana dashboard	Dashboard shows live data within 15 seconds of predictions
IT05	Airflow DAG full pipeline run	All 7 tasks complete successfully (all green)

3. Test Report

3.1 Unit Test Results

Metric	Value
Total Unit Tests	10
Passed	10
Failed	0
Pass Rate	100%
Total Execution Time	3.03 seconds
Test Framework	pytest 7.4.3
Python Version	3.12.3
Date Executed	April 2026

```
• (venv) (base) aminulrony2024@Aminul:~/Desktop/FinPredict$ cd backend/
• (venv) (base) aminulrony2024@Aminul:~/Desktop/FinPredict/backend$ pytest tests/ -v
===== test session starts =====
platform linux -- Python 3.12.3, pytest-7.4.3, pluggy-1.6.0 -- /home/aminulrony2024/Desktop/FinPredict/venv/bin/python3
cachedir: .pytest_cache
rootdir: /home/aminulrony2024/Desktop/FinPredict/backend
plugins: anyio-3.7.1
collected 10 items

tests/test_health.py::test_health_returns_200 PASSED [ 10%]
tests/test_health.py::test_ready_returns_200 PASSED [ 20%]
tests/test_predict.py::test_predict_returns_200 PASSED [ 30%]
tests/test_predict.py::test_predict_response_has_required_fields PASSED [ 40%]
tests/test_predict.py::test_predict_probability_between_0_and_1 PASSED [ 50%]
tests/test_predict.py::test_predict_invalid_payload_returns_422 PASSED [ 60%]
tests/test_preprocessor.py::test_preprocess_returns_dataframe PASSED [ 70%]
tests/test_preprocessor.py::test_preprocess_has_one_row PASSED [ 80%]
tests/test_preprocessor.py::test_debt_to_income_ratio PASSED [ 90%]
tests/test_preprocessor.py::test_gender_encoded_correctly PASSED [100%]
```

3.2 Detailed Unit Test Results

Test ID	Test Name	Result	Duration
TC01	test_health_returns_200	PASSED	0.12s
TC02	test_ready_returns_200	PASSED	0.08s
TC03	test_predict_returns_200	PASSED	1.45s
TC04	test_predict_response_has_required_fields	PASSED	0.31s
TC05	test_predict_probability_between_0_and_1	PASSED	0.28s
TC06	test_predict_invalid_payload_returns_422	PASSED	0.09s
TC07	test_preprocess_returns_dataframe	PASSED	0.22s
TC08	test_preprocess_has_one_row	PASSED	0.18s
TC09	test_debt_to_income_ratio	PASSED	0.17s
TC10	test_gender_encoded_correctly	PASSED	0.13s

3.3 Manual Test Results

Test ID	Description	Result	Notes
MT01	Homepage loads correctly	PASS	All elements render correctly
MT02	Navigate to Apply page	PASS	Routing works correctly
MT03	Form accepts all input types	PASS	Fixed focus loss bug — all inputs stable
MT04	Approved prediction	PASS	78.1% probability, Low risk shown

	displays		
MT05	Rejected prediction displays	PASS	Red card with recommendation shown
MT06	Pipeline page loads	PASS	7 task statuses visible
MT07	User Manual page loads	PASS	All sections visible
MT08	Empty form submission	PASS	Error message appears correctly

3.4 Integration Test Results

Test ID	Description	Result	Notes
IT01	Frontend → Backend prediction	PASS	CORS fixed — OPTIONS 200 OK
IT02	Backend → MLflow model loading	PASS	Model loaded at startup confirmed in logs
IT03	Backend → Prometheus metrics	PASS	Counters increment after predictions
IT04	Prometheus → Grafana dashboard	PASS	Dashboard auto-provisioned, shows live data
IT05	Airflow DAG full pipeline	PASS	All 7 tasks green — 13.4s total
IT06	DVC repro pipeline	PASS	All 7 stages reproduce correctly

3.5 Acceptance Criteria — Final Status

Criteria	Target	Actual	Status
Unit tests pass	10/10	10/10	MET
API HTTP codes correct	200/422/500	Verified	MET
Probability always in range	0.0-1.0	Confirmed	MET
Invalid inputs rejected	422 response	Verified	MET
Inference latency	< 200ms	< 50ms avg	MET
Frontend no console errors	Zero errors	Zero errors	MET
High credit score approved	approved=true	Confirmed	MET
Low credit score rejected	approved=false	Confirmed	MET

Conclusion: All acceptance criteria were met. The system passed 100% of unit tests, all manual tests, and all integration tests. The application is ready for demonstration.